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An Overview of 3GPP Release-15 Study on Enhanced LTE Support for Connected Drones

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Abstract

Cellular connectivity to low altitude unmanned aerial vehicles (UAVs) has received significant interest recently, which has led to a 3GPP study on enhanced LTE support for connected UAVs in Release 15. The objective of the study is to investigate the capability of LTE networks for providing connectivity to UAVs. In this article, we provide an overview of the key findings of the 3GPP Release 15 study item. We first introduce UAV connectivity requirements and performance evaluation scenarios defined in the study. We then discuss radio channel models and the key identified challenges of using LTE networks to provide connectivity to UAVs. We summarize potential solutions to address the challenges, including interference detection and mitigation techniques, mobility enhancements, and UAV identification. Finally, we also shed light on the key features standardized during the Release 15 work item and provide directions for future work in the area of 5G NR.

Introduction

Most of the activities in terms of research and development in the area of mobile broadband have been focused on providing wireless broadband communication to outdoor users on the ground or indoor users in buildings. Recently, providing cellular connectivity to low altitude unmanned aerial vehicles (UAVs, a.k.a. drones) has gathered increasing interest from the industry [1–4], academia [5], and public policy makers [6, 7]. There is a plethora of use cases for supporting aerial vehicles in cellular networks, some of which include search and rescue, surveillance, wildlife conservation, package delivery, and monitoring of critical infrastructure [8].

When aerial vehicles are flying well above eNodeB antennas, they may have a high likelihood of line-of-sight (LoS) propagation conditions to multiple neighboring eNodeBs. In such a scenario, an uplink signal transmitted from an aerial vehicle may become visible and cause interference to multiple neighboring eNodeBs [9]. If this interference is not controlled or mitigated, it may adversely impact the uplink performance of existing users on the ground. Hence, to protect the existing users in the LTE network, the network may have to perform certain actions such as ensuring that the interference is mitigated or performing admission control of aerial vehicles in the network. As a prerequisite to either of these actions, the network may first need to identify the aerial vehicles. With the above mentioned LoS propagation conditions, downlink signals transmitted from multiple eNodeBs may cause downlink interference to the aerial vehicle. Another issue is whether the existing mobility mechanism of LTE networks is sufficient or it needs enhancements to support cellular connectivity of aerial vehicles.

To better answer these issues and to understand the potential of LTE networks for providing cellular connectivity to aerial vehicles, the 3rd Generation Partnership Project (3GPP) started the Release 15 study on enhanced LTE support for aerial vehicles in March 2017 [10]. This study assessed the performance of LTE networks supporting aerial vehicles with up to Release 14 functionality. The study was completed in December 2017, and the outcomes are documented in 3GPP technical report TR 36.777 [1], including comprehensive analysis, evaluation, and field measurement results. With the completion of this study item, 3GPP started a follow-up work item [11] to specify certain features to provide more efficient cellular connectivity to aerial vehicles. The work item on enhanced LTE support for connected aerial vehicles was concluded in June 2018. As of March 2021, preliminary discussions are ongoing on extending the support of UAVs to 5G New Radio (NR) networks.

In this article, we provide an overview of the Release 15 study on enhanced LTE support for aerial vehicles and shed light on the key features standardized during the Release 15 work item. The overview provided by this article is an accessible first reference for researchers interested in learning the 3GPP state-of-the-art findings on cellular connected UAVs. Beyond what is covered in TR 36.777 [1], this article provides in-depth discussions and insights into the UAV-specific solutions that were specified in Release 15. In addition, this article provides evaluation results that highlight the benefits of the solutions specified in Release 15. Finally, we provide directions for future research targeting the support of UAVs in 5G NR networks.

The rest of the article is organized as follows. We provide a description of the UAV-related requirements, scenarios, and channel models. We highlight the uplink and downlink interference problems involving UAVs identified during the Release 15 study. The solutions considered during

Digital Object Identifier:
10.1109/MCOMSTD.0001.1900021

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the study item and those that were standardized in the work item for interference detection are next discussed. We describe interference mitigation techniques and the features studied and standardized in the Release 15 work item. We discuss mobility performance and identified solutions, followed by a description of the features standardized in the Release 15 work item. The identification of UAVs in a cellular network is also discussed. Finally, the article is concluded along with some directions for future research.

Requirements, Scenarios, and Channel Models

Performance Requirements and Deployment Scenarios

The two main data types with regard to wireless connectivity of aerial user equipments (UEs) are command and control data and application data. The ability to send command and control traffic to aerial user equipments (UEs) from eNodeBs can significantly improve the safety and operation of aerial UEs. For instance, it is critical that information such as changes in the flight route are conveyed to the aerial UEs in a timely manner with sufficient reliability. The general performance requirements for UAVs are specified in [15] for UAV altitudes up to 300 m. Depending on specific UAV applications, uplink data rate requirements are in the range of 60 kb/s to 120 Mb/s, and the downlink data rate requirements are in the range of 300 kb/s to 50 Mb/s. Furthermore, the end-to-end latency requirements are in the range of 20 ms to 1 s. For UAV command and control communications, there are reliability requirements in the range of 99.9 to 99.99 percent. In the Release 15 study, uplink/downlink data rate requirement of 60–100 kb/s was defined for command and control traffic to ensure proper operational control of aerial UEs. In addition, the Release 15 study also defined up to 50 Mb/s as uplink data rate requirement for application data.

To evaluate the performance of LTE networks in the presence of LTE connected aerial vehicles, aerial vehicle scenarios including urban-macro (UMa-AV), urban-micro (UMi-AV), and rural-macro (RMa-AV) were considered. UMa-AV represents scenarios where the eNodeB antennas are mounted above the rooftop levels of surrounding buildings in urban environments. Urban scenarios with below-rooftop eNodeB antenna mountings are represented by UMi-AV. Larger cells in rural environments with eNodeB antennas mounted on top of towers are represented by RMa-AV.

In the evaluations during the Release 15 study, aerial vehicles were modeled as outdoor UEs with heights well above ground level, while terrestrial users were on the ground and inside buildings.

Channel Modelling

To characterize the channels between aerial UEs and eNodeBs, the Release 15 study defined models for LoS probability, path loss, shadow fading, and fast fading. In this section, we highlight some key aspects of these models.

To define LoS probability, an aerial UE height-dependent modeling approach was adopted in the study. For all three scenarios, the LoS probability models defined in [12] were reused for aerial UE heights below a lower height threshold. The lower height threshold is 22.5 m for UMa-AV and UMi-AV, while the lower height threshold is 10 m for RMa-AV.

Since eNodeB antennas are well above rooftop in UMa-AV and RMa-AV, a 100 percent LoS probability is assumed above an upper height threshold. The upper height thresholds are 100 m and 40 m for UMa-AV and RMa-AV, respectively. As eNodeB antennas are below rooftop in UMi-AV, the probability of non-line of sight (NLoS) is generally higher in UMi-AV when compared to UMa-AV and RMa-AV. Hence, the upper height threshold is not applicable in UMi-AV.

In the aerial UE height range between the lower and upper height thresholds, the LoS probability models in [1] were derived via ray tracing simulations for UMa-AV and RMa-AV. For UMi-AV, ray tracing simulations were used to derive an LoS probability model that applies for aerial UE heights above the lower height threshold.

For all three scenarios, the path loss and shadow fading models defined in [12] are reused for aerial UE heights below a lower height threshold. The lower height threshold is 22.5 m for UMa-AV and UMi-AV, and 10 m for RMa-AV. In the aerial UE height range above the lower height threshold, path loss and shadow fading models for both LoS and NLoS conditions were agreed in the Release 15 study considering field measurements and ray tracing simulation results contributed by multiple sources.

Problems Identified

Uplink Interference

In the uplink, the aerial UEs were found to cause interference to more cells than a typical terrestrial UE could. This is because aerial UEs, when they are airborne, experience LoS propagation conditions to more cells with higher probability than terrestrial UEs. This generally translates into higher interference caused by an airborne aerial UE to these cells. The uplink interference-over-thermal-noise (IoT) ratios for terrestrial UEs are given in Fig. 1a, which shows the effect of increased uplink interference on terrestrial UEs as the aerial UE ratio increases. For the evaluations in Fig. 1a, a system with 19 macro-sites with 3 sectors per site was assumed. In the uplink, the eNodeB and UE were assumed to be equipped with two receive antennas and one transmit antenna, respectively.

Due to the increased uplink interference, the uplink throughput performance of terrestrial UEs degrades when the aerial UE ratio is increased in the network. The degraded uplink throughput of terrestrial UEs in turn increases the uplink resource utilization level in the network. In other words, degraded uplink throughputs imply that the UEs take longer time to transmit their data, which will consume more resources and lead to increased uplink resource utilization. An increased uplink resource utilization level inherently means an increased level of uplink interference in the network, which in turn was observed to degrade the uplink performance of both aerial UEs and terrestrial UEs.

The two main data types with regard to wireless connectivity of aerial user equipments (UEs) are command and control data and application data. The ability to send command and control traffic to aerial UEs from eNodeBs can significantly improve the safety and operation of aerial UEs.

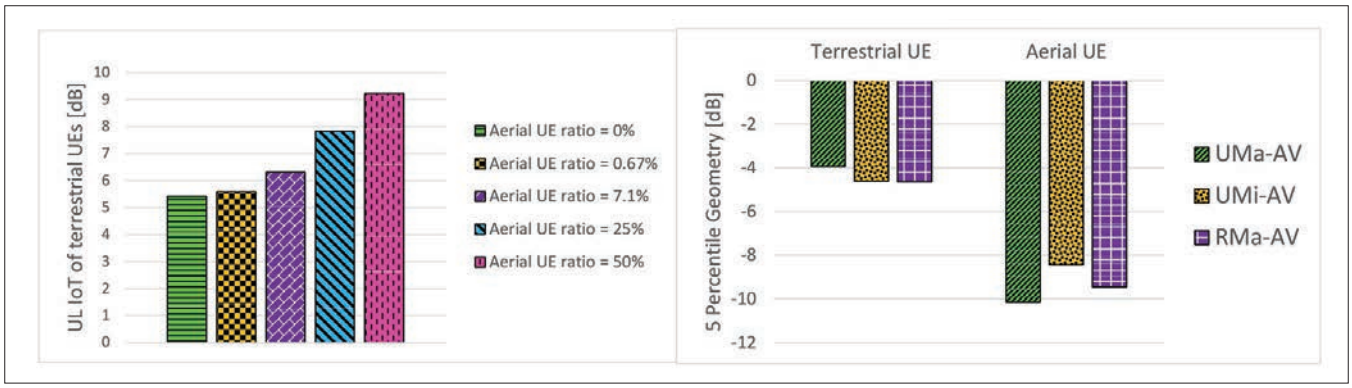

Figure 1. Illustration of UL and DL interference: a) increase in UL IoT of terrestrial UEs with increasing aerial UE ratios; b) comparison of five-percentile geometry showing poorer geometries of aerial UEs when compared to those of terrestrial UEs.

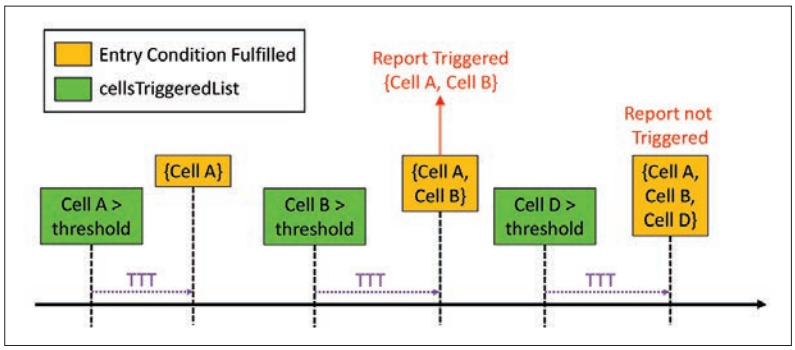

Figure 2. An illustration of UE measurement reporting when a certain preconfigured event is triggered for multiple cells.

Downlink Interference

In the downlink, compared to a typical terrestrial UE, the aerial UEs observe interference from more cells due to the LoS propagation conditions experienced by aerial UEs when they are airborne. Figure 1b compares the five-percentile downlink geometry experienced by aerial UEs when compared to that experienced by terrestrial UEs. For the evaluations in Fig. 1b, a system with 19 macro-sites with 3 sectors per site was assumed. In the downlink, the eNodeB and UE were assumed to be equipped with two transmit antennas and two receive antennas, respectively. Geometry here is defined as the ratio of the average received power from the serving cell to the sum of the average interference power and noise power. The average interference power is computed as the sum of average received power from all non-serving cells. The degraded downlink geometry experienced by the aerial UEs is a result of receiving downlink inter-cell interference from multiple cells.

Due to the increased downlink interference, the downlink throughput performance of aerial UEs degrades. The degraded downlink throughput of aerial UEs when coupled with increased aerial UE ratios increases the downlink resource utilization level in the network. An increased downlink resource utilization level inherently means an increased level of downlink interference in the network, which in turn degrades the downlink performance of both aerial UEs and terrestrial UEs.

To address the challenges due to the uplink and downlink interference, 3GPP studied interference detection and mitigation techniques, mobility enhancements, and UAV identification. Some solutions were standardized to address these challenges during the Release 15 work item, while some issues were concluded to be solvable via implementation-based techniques. In the next sections, we provide an overview and shed light on the solutions standardized during the Release 15 work item.

Solutions for Interference Detection

UE-Based Solutions

In an LTE network, a UE can be configured to perform neighboring cell measurements such as reference signal received power (RSRP), reference signal received quality (RSRQ), and reference signal–signal-to-interference-plus-noise ratio (RS-SINR). Downlink interference can be detected based on these UE measurements reported to the eNodeB. One key aspect is to link the triggering of measurement reports to the changing interference conditions. One of the solutions studied and later specified during the Release 15 work item involves a UE sending a measurement report to the network when the UE sees a certain event criterion being fulfilled simultaneously for multiple cells. In other words, the UE is configured with a triggering condition such that the UE reports a measurement when a certain number of cells X fulfill an entry condition of a given event during a given duration of time, known as a time to trigger (TTT) [14]. This feature is useful for detecting interference from a flying aerial UE. Upon reception of such a measurement report, the network can detect interference early and apply different interference mitigation solutions. For the UE to report measurements when an event criterion is met for multiple cells, the following need to be configured to the UE:

- Number of cells X
- The event to use such as events A3, A4, or A5 as defined in [14]
- The thresholds associated with events

An illustration of UE measurement reporting when an event is triggered for multiple cells is shown in Fig. 2. In this illustration, the number of cells X for which an event must be fulfilled before triggering a measurement report is set to two. When preconfigured events are fulfilled for cells A and B, these cells are added to the

cellsTriggeredList. After both cells A and B are added to the cellsTriggeredList (i.e., corresponding to $X = 2$ cells), a measurement report is triggered from the UE to the network after the TTT duration. It should be noted that when preconfigured events are subsequently fulfilled for more than $X = 2$ cells (e.g., in a third cell, which is cell D in Fig. 2), a measurement report will not be triggered by the UE.

Figure 3 shows a numerical example where the UE is configured to report a measurement when four cells fulfill entry conditions for event A4 as defined in [14]. For the evaluations in Fig. 3, a system with 19 macro-sites with 3 sectors per site was assumed. In this figure, RSRP values corresponding to a set of different cells over time for an aerial UE moving at a speed of 150 km/h with a 200 m altitude are shown. The absolute threshold associated with the A4 event is −76 dBm in the example. In this example, RSRP measurements corresponding to four neighboring cells become better than the threshold of −76 dB near 4.4 s. Hence, the aerial UE will trigger a measurement report shortly after 4.4 s to the network that can be used for interference detection.

A second feature specified during the Release 15 work item is height-based measurement reporting when an aerial UE crosses a height threshold. A height-based measurement report may include the height, the location, and the horizontal/vertical speeds of the aerial UE. The height threshold is configured to the aerial UE via higher-layer signaling. When an aerial UE is flying above a certain height, both the uplink and downlink interference increase. By configuring an aerial UE with an appropriate height threshold, the network can detect interference early when an aerial UE crosses the configured height threshold and sends a height-based report.

Figure 4 shows an example of height-based measurement reporting. In this example, all three aerial UEs are configured with the same height threshold. Aerial UE A, which is moving upward, reports a height-based measurement when it goes above the height threshold. Aerial UE B, which is moving downward, reports a height-based measurement when it goes below the height threshold. Aerial UE C, which is moving horizontally, does not report a height-based measurement as it does not cross the configured height threshold.

Network-Based Solutions

Several network-based interference detection solutions based on information exchange between eNodeBs were considered during the Release 15 study. An example of a solution considered is exchanging uplink reference signal configuration information of aerial UEs. By exchanging this information, a neighboring eNodeB can measure the uplink interference caused by an aerial UE via measuring the power of the uplink reference signal. It should be noted, however, that the feasibility of exchanging this information depends on factors such as the type of backhaul and the feasibility of exchanging such information over a large number of eNodeBs. Furthermore, in Release 15, it was deemed that such solutions can be handled by network implementation, and network-based solutions were not standardized.

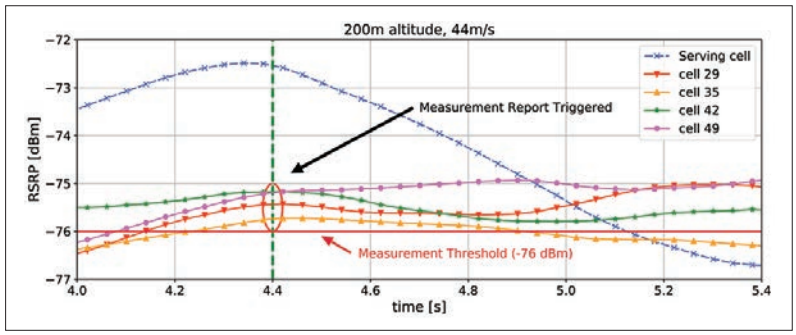

Figure 3. A numerical example for the case where the UE is configured to report a measurement when four cells fulfill entry conditions for event A4.

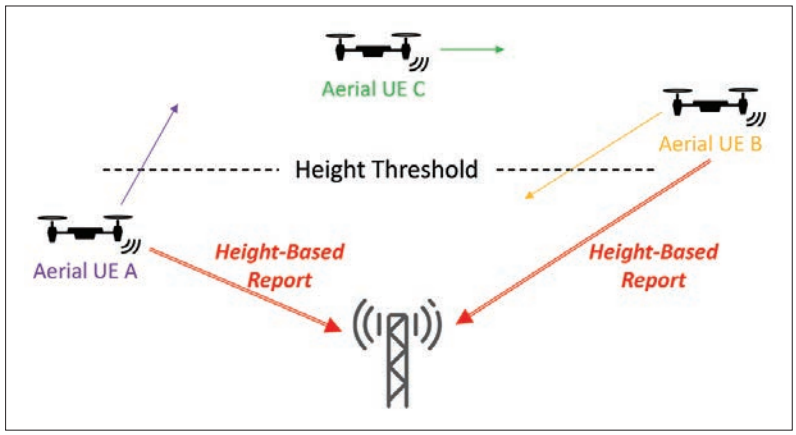

Figure 4. An example illustrating height-based measurement reporting.

Interference Mitigation

Power Control Enhancements for Uplink Interference Mitigation

UE-specific fractional path loss compensation requires the introduction of a UE-specific fractional path loss compensation parameter, which is an enhancement to the open loop power control mechanism that existed in LTE up until Release 14 (see [13, Clause 5.1] for details). With the specification of this technique in the Release 15 work item, the aerial UEs can be configured with a different fractional path loss compensation factor compared to that configured to the terrestrial UEs. Note that depending on the aerial UE height, different aerial UEs can also be configured with different fractional path loss compensation factors. Figure 5 shows the terrestrial UE uplink throughput performance comparison between different combinations of fractional path loss compensation factors configured to the terrestrial and aerial UEs. For these evaluations, a system with 19 macro-sites with 3 sectors per site was assumed. In the uplink, the eNodeB and UE were assumed to be equipped with one receive and one transmit antenna, respectively. In Fig. 5, the path loss compensation factors configured to the terrestrial and aerial UEs are denoted by α_{Terr} and α_{Aer}, respectively. When compared to combination 1, combination 2 can improve terrestrial UEs' 5-percentile uplink throughput by roughly 45 percent and mean uplink throughput by 20 percent. In contrast, combinations 3 and 4 result in significant performance degradation. Hence, the results from Fig. 5 emphasize that the terrestrial UE perfor-

With full-dimensional multiple-input multiple-output (FD-MIMO), multiple transmit antenna ports at the eNodeB can be used to mitigate downlink interference to the aerial UEs. The use of directional antennas at the aerial UE can reduce the downlink interference to the aerial UE from a broad range of angles.

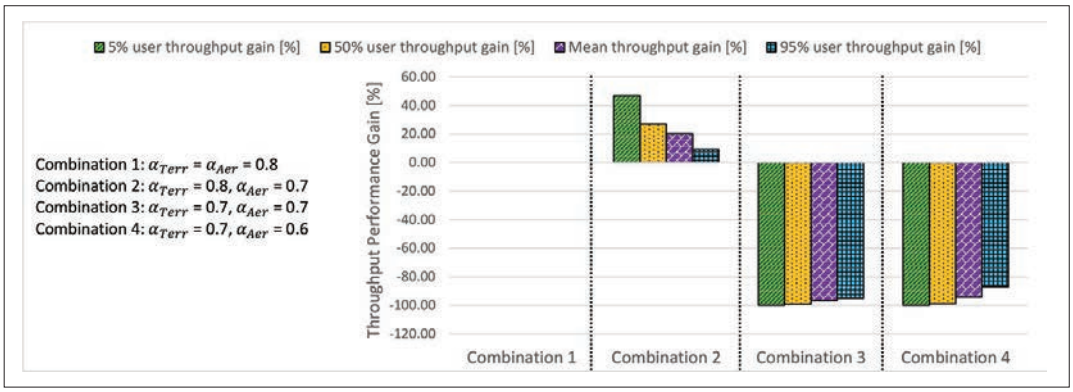

Figure 5. Illustration of uplink throughput performance with the introduction of UE specific fractional path loss compensation parameter in LTE Release 15.

mance can be improved significantly with proper configuration of path loss compensation factors to the aerial and terrestrial UEs, which helps in mitigating uplink interference.

A second power control technique is the use of a UE-specific P_0 parameter (note that P_0 is the target power level that the eNodeB wants to receive per resource block). With this technique, the aerial UEs can be configured with a different P_0 parameter compared to that configured to the terrestrial UEs. Since the P_0 parameter can be UE-specifically configured in LTE, this technique does not require specification enhancements. However, during the Release 15 study, it was found that the range of values supported in LTE for UE-specific P_0 may need to be extended. During the Release 15 work item, the value range of the UE specific component of the P_0 parameter is extended to the range of –16 dB to +15 dB to provide better flexibility in setting open loop power control parameters on a UE specific basis.

In the closed loop power-control-based technique, the target received power of the aerial UEs was adjusted considering measurement reports received from both serving and neighbor cells. Closed loop power control may need to cope with potential fast signal change in the sky. Such fast signal changes are possible in the sky given that the aerial UEs may be served by the sidelobes of eNodeB antennas. To cope with such fast signal changes, the step size of the transmit power control command may need to be increased, which in turn may require specification enhancements to the existing power control mechanism in LTE. However, such a solution was not specified in the Release 15 work item due to lack of time.

Uplink Interference Mitigation based on Directional Antennas

The use of directional antennas at the aerial UE can also help reduce the uplink signal power from the aerial UE in a broad range of angles. This in turn helps reduce the uplink interference caused by the aerial UEs. The following types of aerial UE LoS tracking were considered:

- The antenna direction of the aerial UE is aligned with the direction of travel (DOT) of the UE.
- The LoS direction to the serving cell is either ideally tracked by steering its antenna boresight toward the serving cell or a non-ideally tracked UE with errors due to practical constraints.

As the use of directional antennas at the aerial UE is an implementation issue, there was no specification impact related to this technique in Release 15.

Downlink Interference Mitigation Using FD-MIMO and CoMP

With full-dimensional multiple-input multiple-output (FD-MIMO), multiple transmit antenna ports at the eNodeB can be used to mitigate downlink interference to the aerial UEs. The use of directional antennas at the aerial UE can reduce the downlink interference to the aerial UE from a broad range of angles. When the aerial UEs are equipped with more than two receive antennas, receive beamforming can be an effective interference mitigation technique in the downlink. In the intra-site joint transmission coordinated multipoint (JT CoMP) scheme, data are jointly transmitted to the UEs from multiple cells that belong to the same site. Coverage extension techniques can also be used to enhance synchronization and initial access performance of aerial UEs.

The above downlink interference mitigation solutions studied during Release 15 were either already supported in LTE (FD-MIMO, intra-site JT CoMP, and coverage extension) or implementation-based techniques (directional antennas at the aerial UE and receive beamforming). Hence, it was concluded that these studied solutions were sufficient to guarantee reliable downlink interference mitigation, and further downlink interference mitigation features were not introduced during the Release 15 work item.

Mobility and Aerial UE Identification

Mobility Performance and Solutions Identified

During the Release 15 study, mobility simulations were performed, and measurements from field trials were collected [1]. From these results, the mobility performance of an aerial UE is shown to be worse when compared to that of a terrestrial UE, especially when the number of aerial UEs is large. Due to the increased downlink interference, the downlink SINR for the aerial UEs is much worse than the downlink SINR for the terrestrial UEs. Hence, the aerial UEs may experience more

handover failures, more radio link failures, longer handover interruption time, and so on. The mobility simulation results showed better mobility performance for aerial UEs in the RMa-AV scenario than in the UMa-AV scenario. It should be noted, however, that the interference mitigation techniques listed above were not considered in the mobility simulations of the Release 15 study, and the use of such techniques is expected to improve the aerial UEs' mobility performance.

Some techniques for improving mobility performance of aerial UEs include enhancements to handover procedure such as conditional handover and handover-related parameters considering, for example, location information, airborne status, and flight path plans, and enhancements to existing measurement reporting mechanisms including new events, enhancements of triggering conditions, and so on.

Flight Path Signaling

During the work item of Release 15, a flight path reporting feature was specified. In this feature, the network can request flight path information from the aerial UE at any time, and the UE signals the flight path information if it is available at the UE. In order to limit unnecessary signaling, an aerial UE can indicate to the network if it has flight path information available (e.g., when radio resource control connection is initiated). Upon receiving the flight path information request, the aerial UE reports the flight path information to the network. Figure 6 shows an example of flight path information reporting. As shown in the figure, the flight path information includes the waypoints (p_1, p_2, ...) and the corresponding time stamps (t_1, t_2, ...). The flight path plan can be used by the network to know, for instance, how many UAVs are to be served in an area. This may help the network to plan the used resources.

Aerial UE Identification

Depending on country-specific regulations, an aerial UE may need to be identified by the network in order to allow the use of LTE networks for aerial UE connectivity. Another aspect is that there may be drone-specific service or charging by the operator.

Aerial UE identification can be a combination of user-based identification via subscription information and device-functionality-based identification via LTE radio capability signaling. The mobility management entity (MME) can signal the subscription information to the eNodeB, which can include information on whether the user is authorized to operate for aerial usage. In addition, an aerial UE as an LTE device can indicate its support of aerial-related functions via radio capability signaling to the eNodeB. The combination of the subscription information and the radio capability indication from the UE can be used by the eNodeB to identify an aerial UE, and then perform the necessary control and the relevant functions.

Conclusions and Future Directions

The Release 15 study on LTE connected UAVs concluded that it is feasible to use existing LTE networks to provide connectivity to low altitude UAVs despite some challenges, as overviewed in this article. The article also sheds light on the features that were specified during the Release 15 work item. The features were specified to enhance interference detection, uplink interference mitigation, mobility performance, and aerial UE identification. This article has particularly focused on the 3GPP state-of-the-art findings and solutions on LTE connected UAVs.

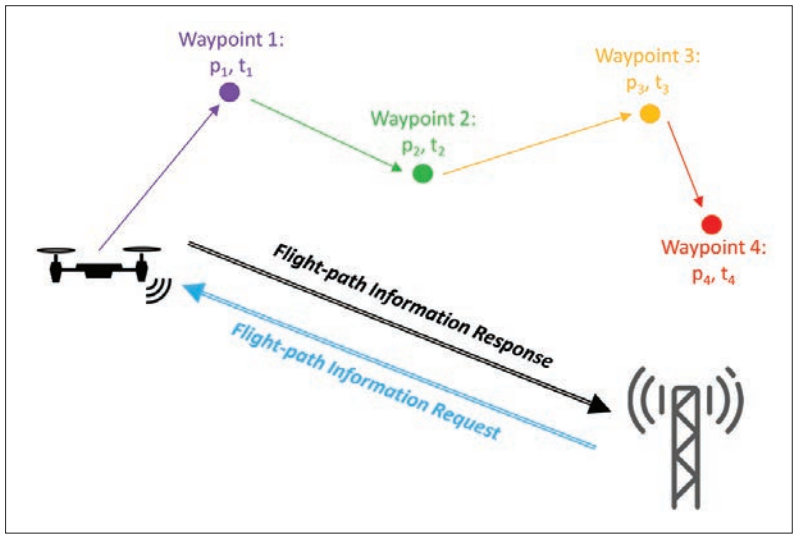

Figure 6. An example illustrating flight path plan reporting.

With the uptake of 5G NR around the world, the support of 5G connected UAVs is one important direction for further research and standardization. The key characteristics of 5G NR include ultra-lean beam-based transmissions, support for ultra-reliable low-latency communications, and spectrum flexibility. From the 3GPP services and systems aspects perspective, studies related to remote identification of unmanned aerial systems (UASs) and application layer support for UASs were performed in Releases 16 and 17, respectively. In Release 17, requirements related to key performance indicators for communication services and command and control traffic, and onboard radio access nodes were specified [15]. However, from the 3GPP radio access network perspective, there has not been any study or specification work related to support of UAVs in 5G NR in Releases 16 and 17. Hence, in Release 18, there is strong interest in specifying features to support UAVs in 5G NR networks. Some key open directions for future research are as follows:

- The first step in evolving 5G NR toward supporting UAVs is to inherit the features introduced in LTE in Release 15. However, since 5G NR has a beam-centric design, how this can be leveraged to improve coverage in the sky is a key open issue that needs to be addressed.
- A 5G NR enabled drone may be in RRC IDLE mode as a drone may be in automated flight. Hence, how to support idle mode mobility for UAVs in 5G NR is a second open issue that needs to be studied. For example, height thresholds may be broadcast by the 5G NR network, which may be used by the UAVs to define the flying modes while they are in RRC IDLE. This enables the UAVs to determine the flying

mode and inform the UE's flying mode early to the network when establishing a connection with the 5G NR network.

- In some scenarios, the UAVs may need to be positioned with very high accuracy. However, positioning accuracy of UAVs in the sky was not investigated during the Release 15 study. Hence, achievable positioning accuracy in the sky for UAVs and potential improvements are interesting topics to be investigated.
- The height of the aerial vehicles was limited up to 300 m in the Release 15 study. However, there may be 5G NR use cases that require higher altitude aircraft with heights much beyond 300 m to be supported. The achievable coverage, reliability, and throughput need to be further studied to identify potential solutions.

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Biographies

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